Domain decomposition for neural networks

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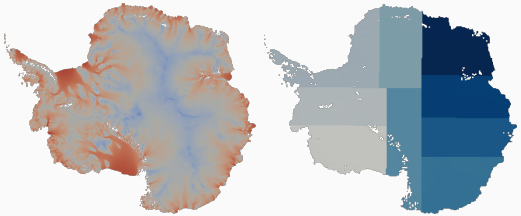
Domain Decomposition Methods



Images based on **Heinlein, Perego, Rajamanickam (2022)**

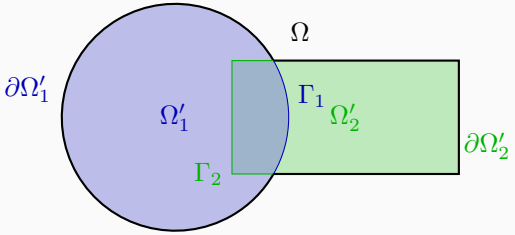
Idea

Decomposing a large **global problem** into smaller **local problems**:

- **Better robustness** and **scalability** of numerical solvers
- **Improved computational efficiency**
- Introduce **parallelism**

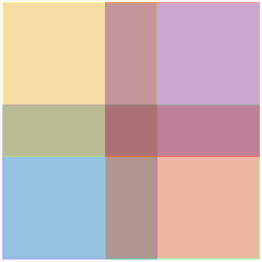
Historical remarks: The **alternating Schwarz method** is the earliest **domain decomposition method (DDM)**, which has been invented by **H. A. Schwarz** and published in **1870**:

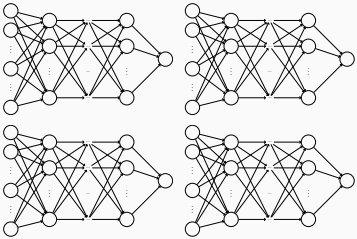
- Schwarz used the algorithm to establish the **existence of harmonic functions** with prescribed boundary values on **regions with non-smooth boundaries**.

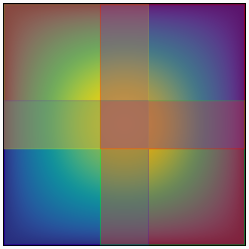


Domain Decomposition for Neural Networks

I)







II)

Domain Decomposition Methods and Machine Learning – Literature

A **non-exhaustive literature overview**:

- **Machine Learning for adaptive BDDC, FETI–DP, and AGDSW**: **Heinlein, Klawonn, Lanser, Weber (2019, 2020, 2021, 2021, 2021, 2022); Klawonn, Lanser, Weber (2024)**
- **cPINNs, XPINNs**: **Jagtap, Kharazmi, Karniadakis (2020); Jagtap, Karniadakis (2020)**
- **Classical Schwarz iteration for PINNs oret DeepRitz (D3M, DeepDDM, etc)**: **Li, Tang, Wu, and Liao (2019); Li, Xiang, Xu (2020); Mercier, Gratton, Boudier (arXiv 2021); Li, Wang, Cui, Xiang, Xu (2023); Sun, Xu, Yi (arXiv 2022, arXiv 2023); Kim, Yang (2022, arXiv 2023)**
- **FBPINNs**: **Moseley, Markham, and Nissen-Meyer (2023); Dolean, Heinlein, Mishra, Moseley (2024, acc. 2024 / arXiv:2306.05486); Heinlein, Howard, Beecroft, Stinis (subm. 2024 / arXiv:2401.07888)**
- **DDMs for CNNs**: **Gu, Zhang, Liu, Cai (2022); Lee, Park, Lee (2022); Klawonn, Lanser, Weber (acc. 2024); Verburg, Heinlein, Cyr (in prep.)**

An overview of the state-of-the-art in **early 2021**:

A. Heinlein, A. Klawonn, M. Lanser, J. Weber

Combining machine learning and domain decomposition methods for the solution of partial differential equations — A review

GAMM-Mitteilungen. 2021.

An overview of the state-of-the-art in the **end of 2023**:

A. Klawonn, M. Lanser, J. Weber

Machine learning and domain decomposition methods — A survey

arXiv:2312.14050. 2023

Outline

1. FBPINNs – Domain Decomposition for Physics-Informed Neural Networks

   Based on joint work with

   Victorita Dolean (TU Eindhoven)

   Ben Moseley and **Siddhartha Mishra** (ETH Zürich)

2. DDU-Net – Domain Decomposition for Convolutional Neural Networks

   Based on joint work with

   Eric Cyr (Sandia National Laboratories)

   Corné Verburg (Delft University of Technology)

FBPINNs – Domain Decomposition for Physics-Informed Neural Networks

Physics-Informed Neural Networks (PINNs)

In the **physics-informed neural network (PINN)** approach introduced by **Raissi et al. (2019)**, a **neural network** is employed to **discretize a partial differential equation**

$$\mathcal{N}[u] = f, \quad \text{in } \Omega.$$

PINNs use a **hybrid loss function**:

$$\mathcal{L}(\boldsymbol{\theta}) = \omega_{\text{data}}\mathcal{L}_{\text{data}}(\boldsymbol{\theta}) + \omega_{\text{PDE}}\mathcal{L}_{\text{PDE}}(\boldsymbol{\theta}),$$

where ω_{data} and ω_{PDE} are **weights** and

$$\mathcal{L}_{\text{data}}(\boldsymbol{\theta}) = \frac{1}{N_{\text{data}}} \sum_{i=1}^{N_{\text{data}}} \left(u(\hat{\boldsymbol{x}}_i, \boldsymbol{\theta}) - u_i\right)^2,$$

$$\mathcal{L}_{\text{PDE}}(\boldsymbol{\theta}) = \frac{1}{N_{\text{PDE}}} \sum_{i=1}^{N_{\text{PDE}}} \left(\mathcal{N}[u](\boldsymbol{x}_i, \boldsymbol{\theta}) - f(\boldsymbol{x}_i)\right)^2.$$

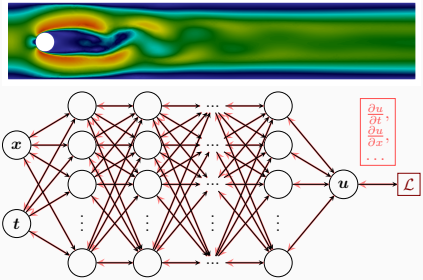
See also **Dissanayake and Phan-Thien (1994); Lagaris et al. (1998)**.

Advantages

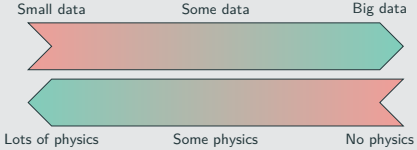
- **"Meshfree"**
- **Small data**
- **Generalization properties**
- **High-dimensional problems**
- **Inverse** and **parameterized problems**

Drawbacks

- **Training cost** and **robustness**
- **Convergence not well-understood**
- **Difficulties with scalability** and **multi-scale problems**



Hybrid loss



- **Known solution values** can be included in $\mathcal{L}_{\text{data}}$
- **Initial and boundary conditions** are also included in $\mathcal{L}_{\text{data}}$

Motivation – Some Observations on the Performance of PINNs

Solve

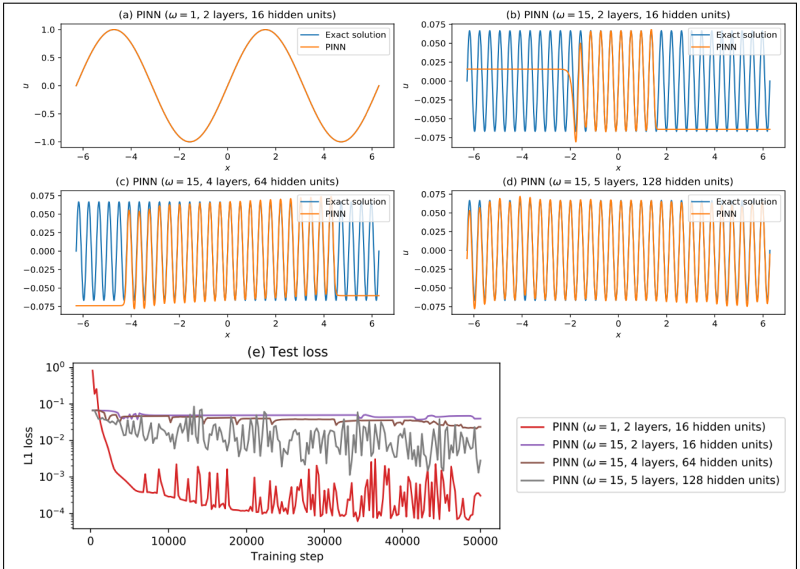
$$\begin{aligned} u' &= \cos(\omega x), \\ u(0) &= 0, \end{aligned}$$

for different values of ω using **PINNs with varying network capacities**.

Scaling issues

- Large computational domains
- Small frequencies

Cf. **Moseley, Markham, and Nissen-Meyer (2023)**



(a) 321 free parameters (d) 66 433 free parameters

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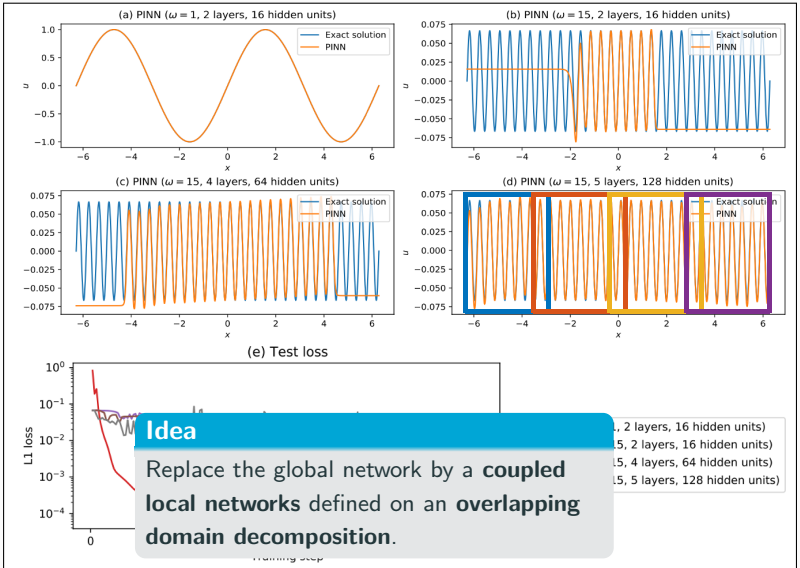


Idea

Replace the global network by a **coupled local networks** defined on an **overlapping domain decomposition**.

(a) 321 free parameters (d) 66 433 free parameters

Finite Basis Physics-Informed Neural Networks (FBPINNs)

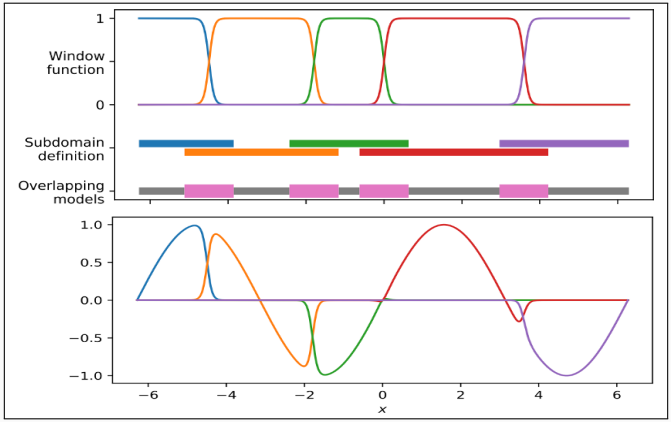
Finite basis physics informed neural networks (FBPINNs) introduced in **Moseley, Markham, and Nissen-Meyer (2023)** use the **network architecture**

$$u(\theta_1, \dots, \theta_J) = \mathcal{C} \sum_{j=1}^{J} \omega_j u_j(\theta_j)$$

and the **loss function**

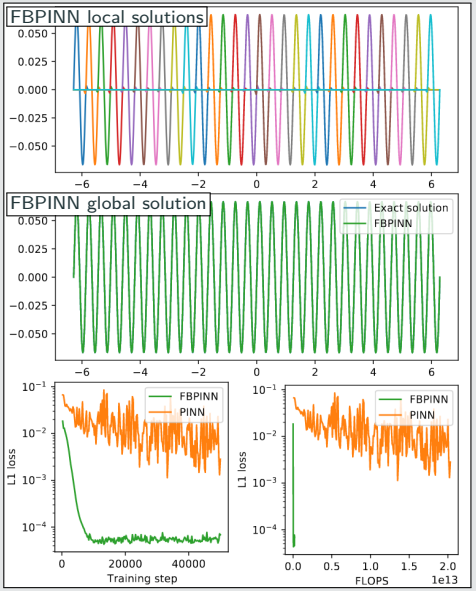
$$\mathcal{L}(\theta_1, \dots, \theta_J) = \frac{1}{N} \sum_{i=1}^{N} \left(\mathcal{N}[\mathcal{C} \sum_{x_i \in \Omega_j} \omega_j u_j](x_i, \theta_j) - f(x_i) \right)^2.$$

- **Window functions**: ω_j with $\text{supp}(\omega_j) \subset \Omega_j$ and $\sum_{j=1}^{J} \omega_j \equiv 1$ on Ω
- **Overlapping DD**: $\Omega = \bigcup_{l=1}^{J} \Omega_j$



Numerical Results for FBPINNs

PINN vs FBPINN (Moseley et al. (2023))



Scalability of FBPINNs

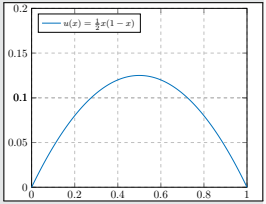
Consider the **simple boundary value problem**

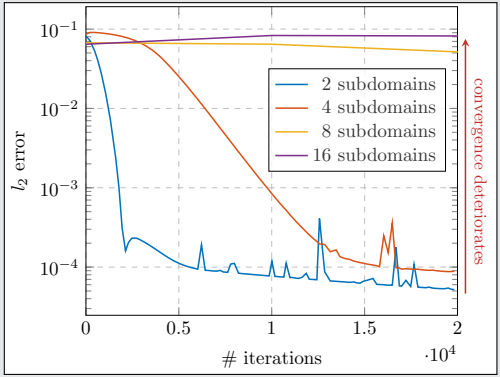
$$-u'' = 1 \text{ in } [0,1],$$

$$u(0) = u(1) = 0,$$

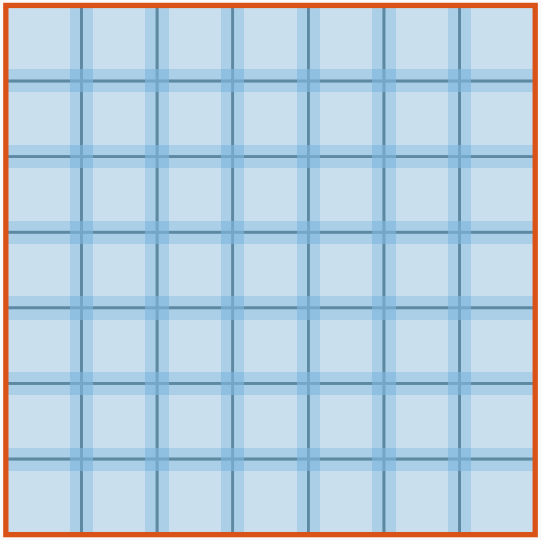
which has the **solution**

$$u(x) = 1/2 x(1-x).$$





Transport of Information One-Level Overlapping Schwarz Methods



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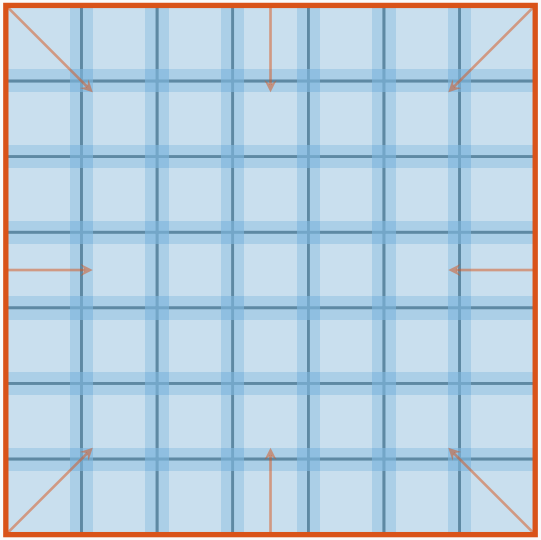

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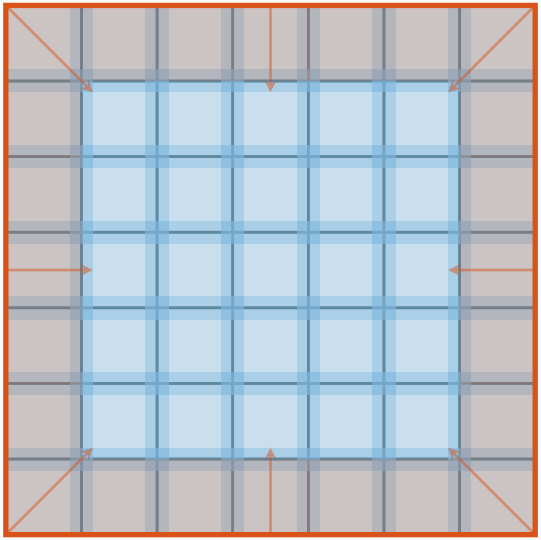

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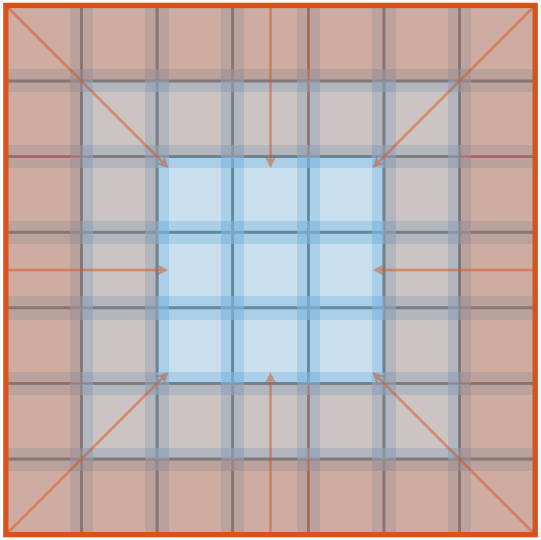

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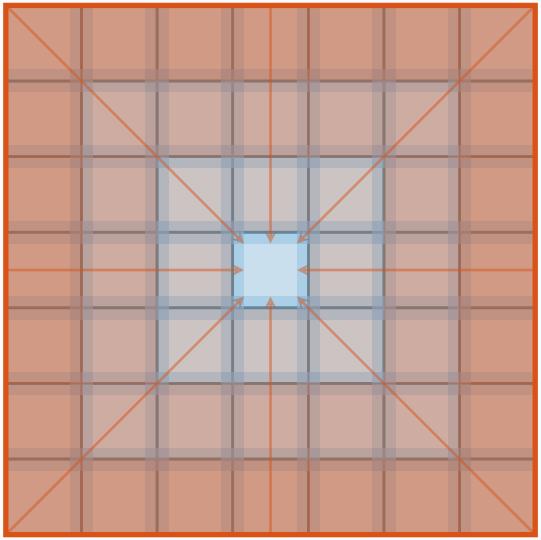

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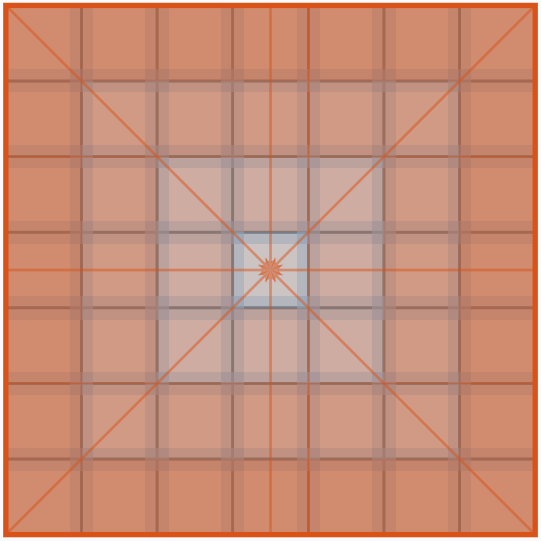
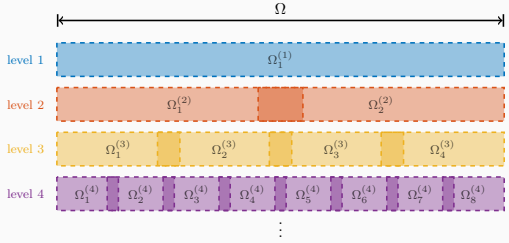


Information (in particular, boundary data) is **only exchanged via the overlapping regions**, leading to **slow convergence** $\rightarrow$ establish a **faster / global transport of information**.

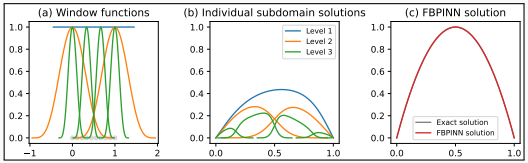
Multi-Level FBPINN Algorithm

Extension of FBPINNs to L levels; Cf. **Dolean, Heinlein, Mishra, Moseley (accepted 2024 / arXiv:2306.05486)**.



L-level network architecture

$$u\left(\theta_1^{(1)}, \ldots, \theta_{J(L)}^{(L)}\right) = \mathcal{C}\left(\sum_{l=1}^{L} \sum_{i=1}^{N^{(l)}} \omega_j^{(l)} u_j^{(l)}\left(\theta_j^{(l)}\right)\right)$$



Multi-Frequency Problem

Let us now consider the **two-dimensional multi-frequency Laplace boundary value problem**

$$-\Delta u = 2 \sum_{i=1}^{n} (\omega_i \pi)^2 \sin(\omega_i \pi x) \sin(\omega_i \pi y) \quad \text{in } \Omega,$$

$$u = 0 \quad \text{on } \partial\Omega,$$

with $\omega_i = 2^i$.

For increasing values of n, we obtain the **analytical solutions**:

Multi-Level FBPINN Algorithm

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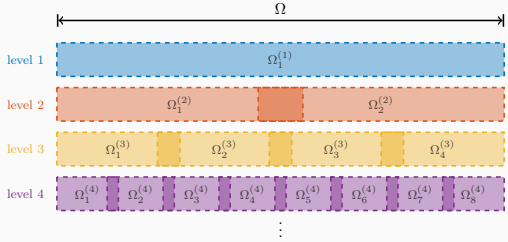

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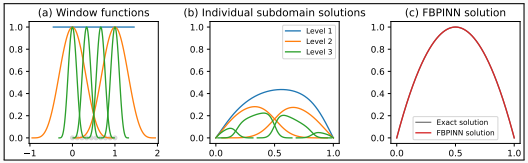

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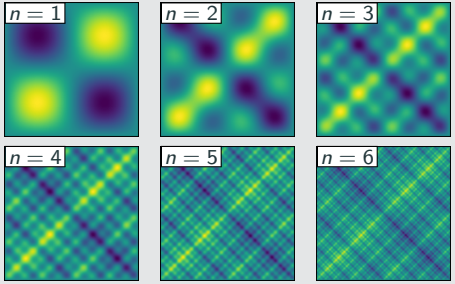
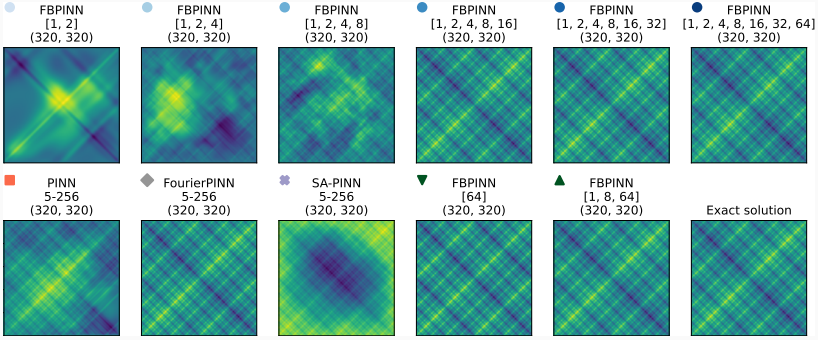
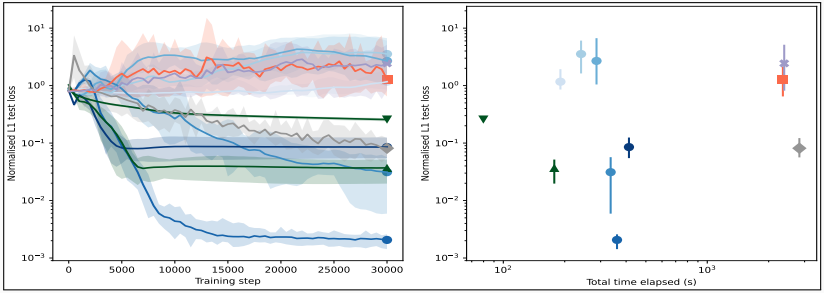
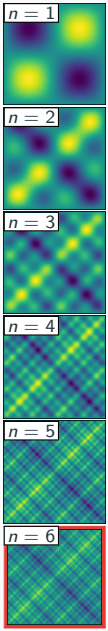


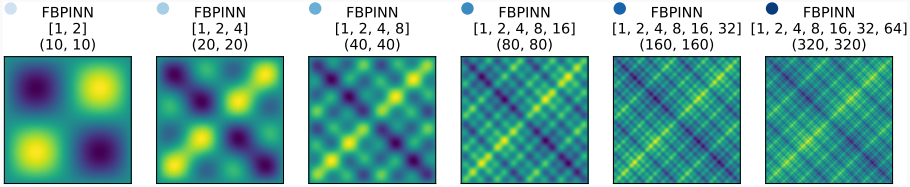
Multi-Level FBPINNs for a Multi-Frequency Problem – Strong Scaling

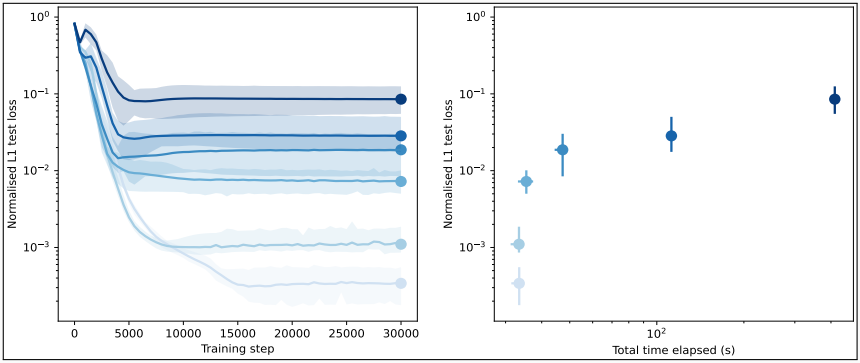


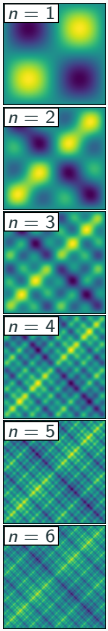




Multi-Level FBPINNs for a Multi-Frequency Problem – Weak Scaling





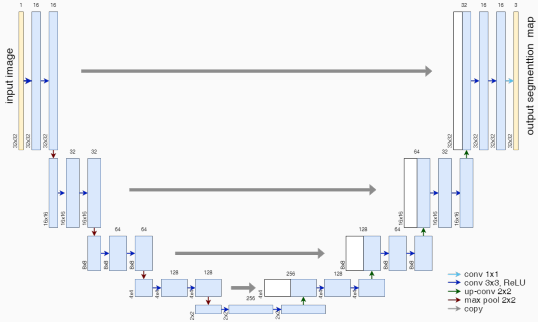


- Ongoing: **analysis and improvement of the convergence**

Cf. **Dolean, Heinlein, Mishra, Moseley (accepted 2024 / arXiv:2306.05486)**. Implementation using JAX

DDU-Net – Domain Decomposition for Convolutional Neural Networks

Memory Requirements for CNN Training

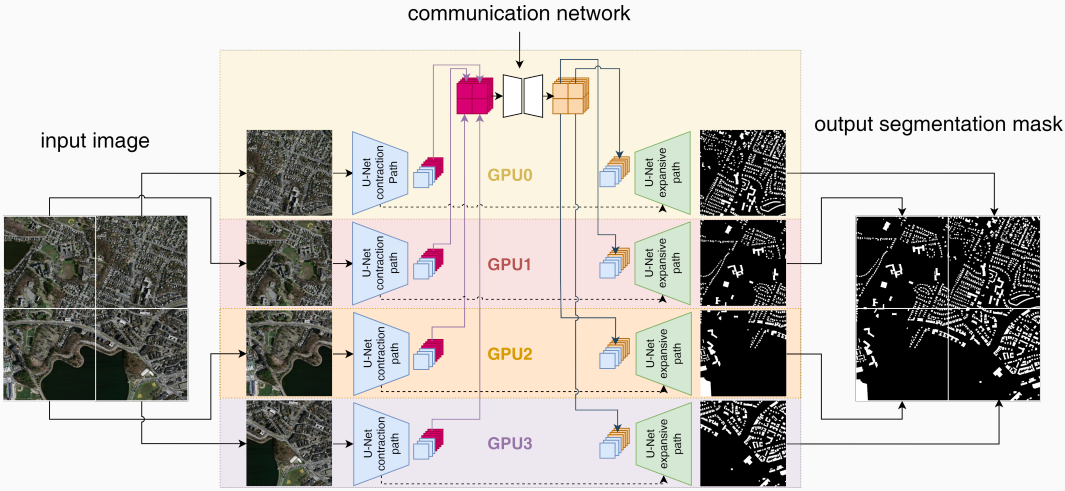


- As an example for a **convolutional neural network (CNN)**, we employ the **U-Net architecture** introduced in **Ronneberger, Fischer, and Brox (2015)**.
- The U-Net yields **state-of-the-art accuracy in semantic image segmentation** and other **image-to-image tasks**.

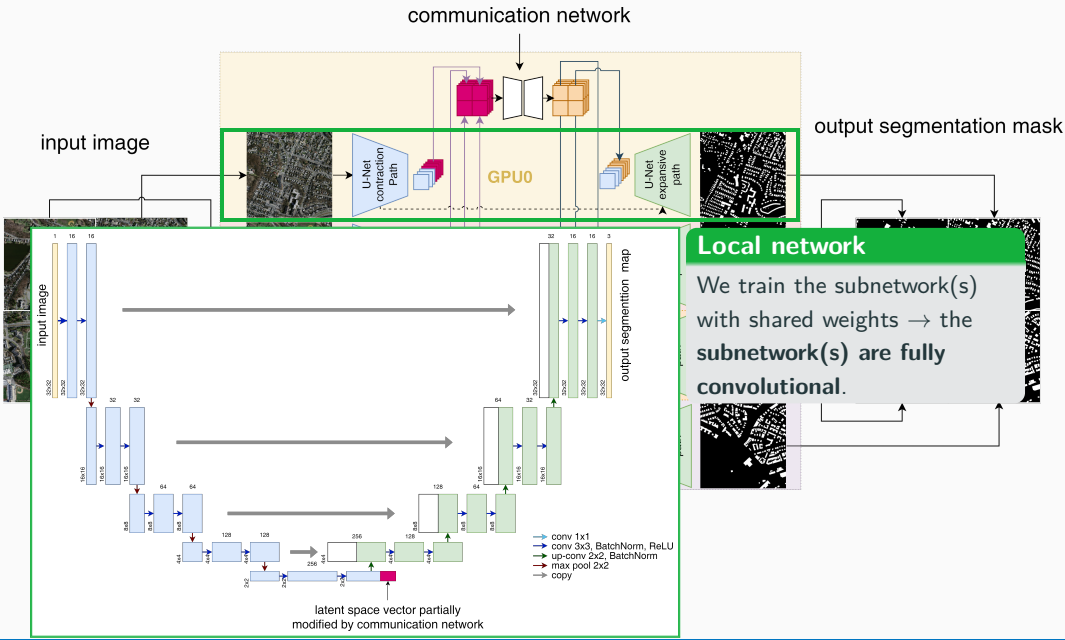
Below: *memory consumption for training on a single* 1024×1024 *image.*

name	size	# channels		mem. feature maps		mem. weights	
		input	output	# of values	MB	# of values	MB
input block	1 024	3	64	268 M	**1 024.0**	38 848	**0.148**
encoder block 1	512	64	128	167 M	**704.0**	221 696	**0.846**
encoder block 2	256	128	256	84 M	**352.0**	885 760	**3.379**
encoder block 3	128	256	512	42 M	**176.0**	3 540 992	**13.508**
encoder block 4	64	512	1 024	21 M	**88.0**	14 159 872	**54.016**
decoder block 1	64	1,024	512	50 M	**192.0**	9 177 088	**35.008**
decoder block 2	128	512	256	101 M	**384.0**	2 294 784	**8.754**
decoder block 3	256	256	128	201 M	**768.0**	573 952	**2.189**
decoder block 4	512	128	64	402 M	**1 536.0**	143 616	**0.548**
output block	1 024	64	3	3.1 M	**12.0**	195	**0.001**

Decomposing the U-Net



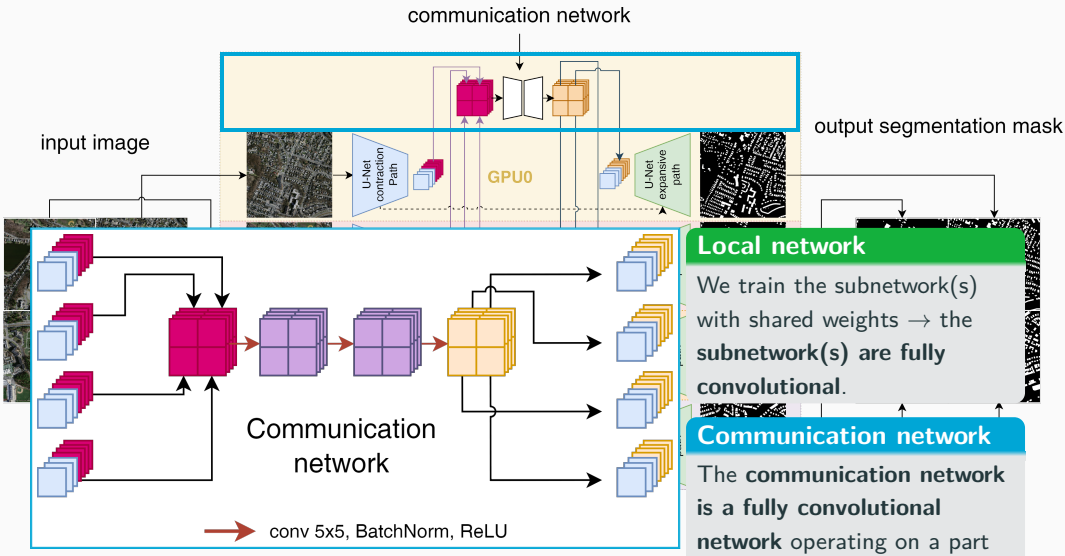
Decomposing the U-Net



Local network

We train the subnetwork(s) with shared weights → the **subnetwork(s) are fully convolutional**.

Decomposing the U-Net



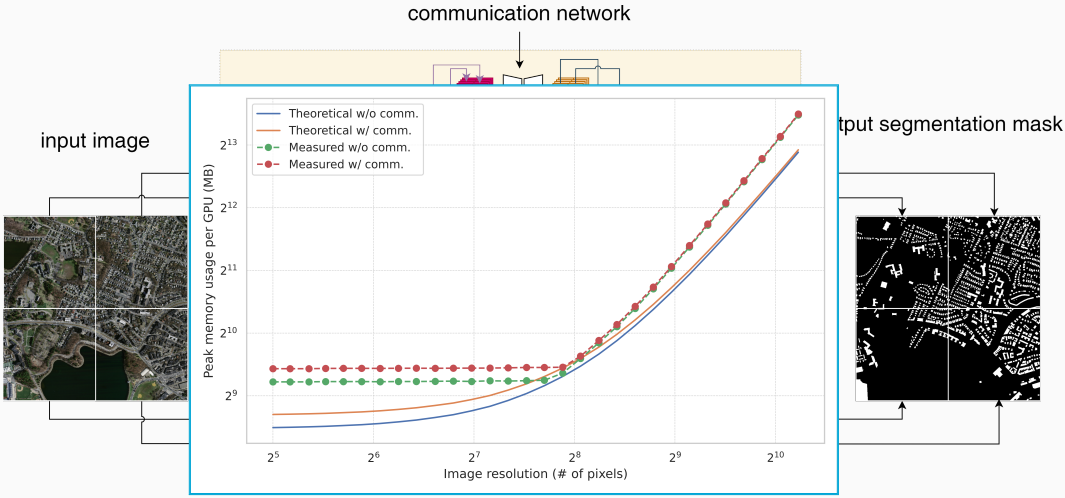
Local network

We train the subnetwork(s) with shared weights $\rightarrow$ the **subnetwork(s) are fully convolutional**.

Communication network

The **communication network is a fully convolutional network** operating on a part of the coarse feature maps.

Decomposing the U-Net

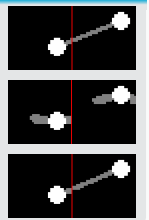


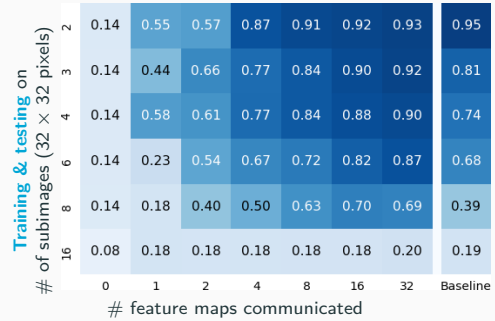
- Distribution of feature maps results in **significant reduction of memory usage on a single GPU**
- **Moderate additional memory usage** due to the **communication network**

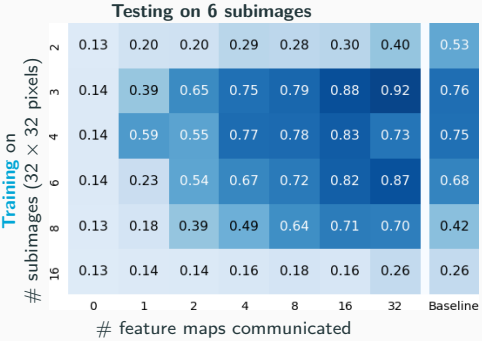
Results – Synthetic Data Set

Task: Connect two dots via a line segment

Result: Communication



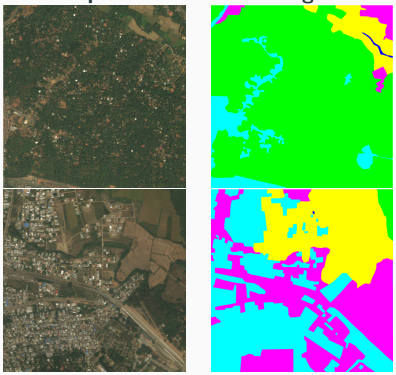




DeepGlobe 2018 Satellite Image Data Set (Demir et al. (2018))

class	pixel count	proportion
urban	642.4M	9.35 %
agriculture	3898.0M	56.76 %
rangeland	701.1M	10.21 %
forest	944.4M	13.75 %
water	256.9M	3.74 %
barren	421.8M	6.14 %
unknown	3.0M	0.04 %

Input Target

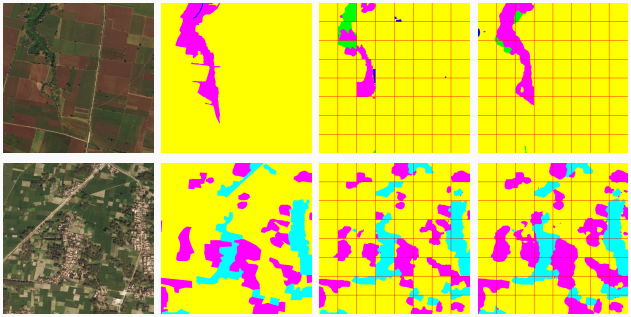


Avoiding overfitting

The data set includes **only 803 images**. To **avoid overfitting**, we

- apply **batch normalization** and use **random dropout** layers and **data augmentation**
- **initialize the encoder** using the **ResNet-18** (**He, Zhang, Ren, and Sun(2016)**)

image true mask without comm. with comm.



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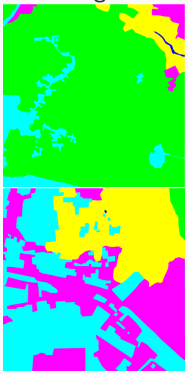
Input

Target



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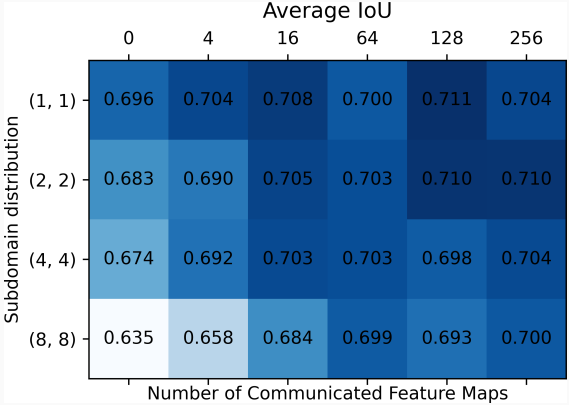

FBPINNs – Domain Decomposition for Physics-Informed Neural Networks

- **Training of PINNs** can be **challenging**.
- **Schwarz domain decomposition architectures improve the scalability of PINNs** to **large domains** / **high frequencies**, **keeping the complexity of the local networks low**.
- As classical domain decomposition methods, **one-level FBPINNs** are **not scalable to large numbers of subdomains**; **multilevel FBPINNs enable scalability**.

→ Friday, MS132A: **P. Stinis. Multifidelity scientific machine learning**

DDU-Net – Domain Decomposition for CNNs

- The **memory requirements for training of high-resolution images** using CNNs can be **large**, In particular, the U-Net model requires **storing intermediate feature maps**.
- The **decomposition of the images in the training of the U-Net** can help to **distribute the memory load** among **multiple GPUs**.
- Our **novel DDU-Net** approach **decouples the training on the sub-images**, resulting in a loss of communication. In order to enable **moderate communication**, a **communication network** is introduced at the deepest level of the U-Net architecture.

Thank you for your attention!